

PTv3-Drone-Flow: Action-Conditioned 3D Scene Flow Prediction for Drone Navigation Using Point Transformer V3

Ray Chen Qingyong Meng Jiajin Li

Abstract

We present **PTv3-Drone-Flow**, an end-to-end pipeline for action-conditioned 3D scene flow prediction applied to drone navigation. Building on Point Transformer V3 (PTv3) [6] and PointWorld [2], we adapt the 3D world model paradigm from robot arm manipulation to aerial navigation. Given a drone’s RGB-D observation and odometry action, our model predicts ego-motion induced 3D point flow. We train on 822 real-world samples collected from a drone flight using Intel RealSense and NVIDIA nvblox, achieving a validation end-point error (EPE) of **0.793m** and MSE loss of **0.255** over 25 epochs. Our work demonstrates the feasibility of PTv3-based world models for drone scene understanding and provides a reproducible open-source pipeline from ROS2 bag files to a trained model.

1. Introduction

Predicting how a 3D scene evolves as a robot moves through it is a fundamental challenge in embodied AI. For drone navigation, this capability enables anticipatory obstacle avoidance, depth-aware path planning, and closed-loop control without relying on GPS or pre-built maps.

Scene flow — the 3D motion field of points in a scene — provides a compact representation of this dynamic information. While classical methods relied on multi-view stereo and optical flow [5], recent deep learning approaches have demonstrated strong performance on point cloud data [4]. We ask: *can a PTv3-based world model trained on real drone data predict ego-motion induced 3D scene flow?*

Our contributions: (1) an end-to-end pipeline converting ROS2 bag files to WebDataset format for PTv3 training; (2) a proof-of-concept applying PTv3-based world models to drone ego-motion scene flow; (3) a real-world evaluation on 822 samples from an indoor drone flight; (4) open-source: <https://github.com/RayChen666/PTv3-drone-flow>.

2. Related Work

PointNet [4] processes each point independently via a shared MLP with global max pooling, achieving permutation invariance. However, treating points in isolation regardless of spatial proximity fundamentally limits its fine-grained 3D scene understanding.

PointLLM [7] bridges point cloud understanding with Large Language Models via language grounding, reflecting the broader shift from task-specific 3D recognition toward foundation model-based perception — contextualizing where the field is heading beyond traditional scene flow methods.

PointWorld [2] extends 3D perception to temporal dynamics, predicting how a scene evolves given a robot action. Its use of PTv3 as a spatially-aware scene encoder inspires our backbone choice. However, it is limited to robot arm manipulation related tasks. We extend this paradigm from robot arm manipulation to aerial drone navigation with 822 real-world data samples.

3. Method

3.1 Formulation Given a point cloud $\mathbf{P}_0 \in \mathbb{R}^{N \times 3}$ at time t and a drone action $\mathbf{a} \in \mathbb{R}^{13}$ (pose delta + velocity), we predict the scene flow $\hat{\mathbf{F}} \in \mathbb{R}^{N \times 3}$ such that $\mathbf{P}_1 \approx \mathbf{P}_0 + \hat{\mathbf{F}}$.

3.2 Data Pipeline We extract depth images from `/race10/stereo/img_depth`, and odometry from `/race10/odom`, based on a ROS2 bag file recorded during an indoor drone flight. Depth images are back-projected to 3D point clouds using camera intrinsics. To ensure meaningful motion signal, we pair frames with a stride of 10 ($\approx 0.83s$ at 12Hz), yielding 822 training samples stored as WebDataset shards. Each sample is normalized to zero mean and unit scale.

3.3 Model Architecture Our model has three components: (a) **PTv3 Encoder** — PTv3-small (29.6M parameters) with 5 encoder stages, serialized attention via z-order and Hilbert curve orderings, and sparse convolution, taking flattened point coordinates

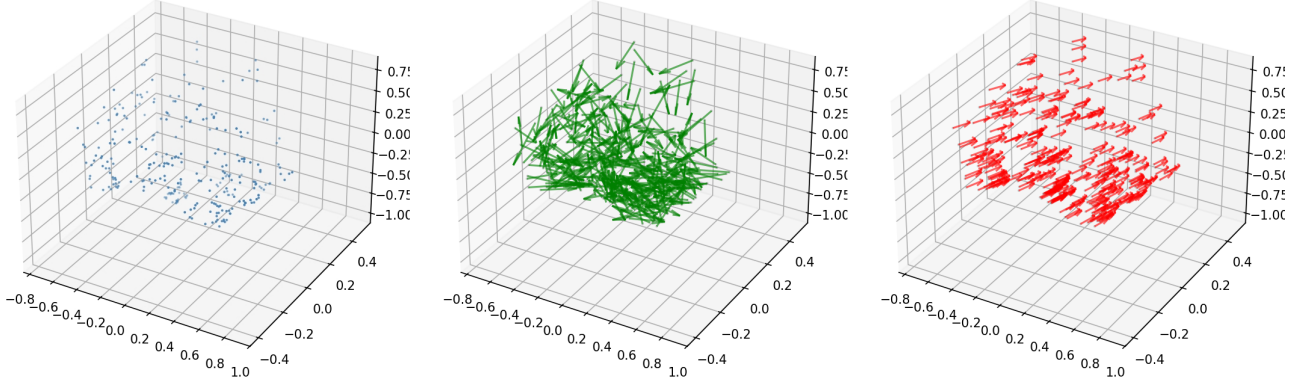


Figure 1. Flow visualization. Left: input point cloud (blue). Middle: ground truth flow (green). Right: PTv3 predicted flow (red).

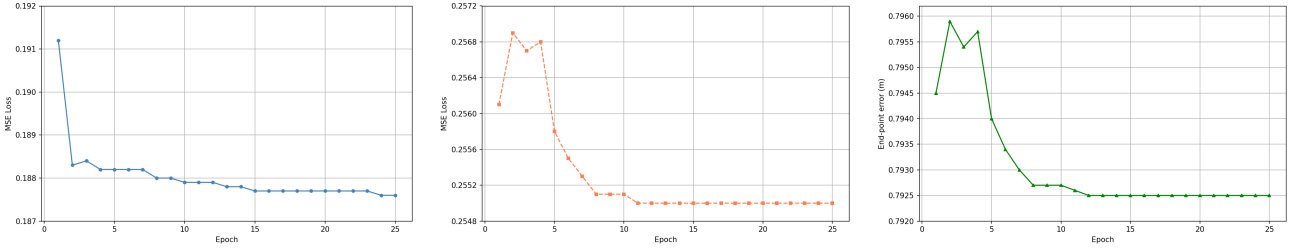


Figure 2. Training curves over 25 epochs. Left: Train MSE loss. Middle: Validation MSE loss. Right: Validation EPE (m).

as input features; **(b) Action Embedding** — the 13-dimensional action vector projected to bottleneck dimension (384) via a two-layer GELU MLP; and **(c) Flow Head** — global average pooling over coarsest encoder features, conditioned by action embedding via addition, passed through a three-layer MLP to produce a global flow vector broadcast to all N points.

3.4 Training We train with AdamW ($\text{lr}=10^{-3}$, weight decay= 10^{-4}) and cosine annealing scheduler for 25 epochs. Training is conducted on a cloud-hosted NVIDIA A10 GPU (24GB VRAM) running Ubuntu 22.04, with CUDA 12.1 and PyTorch 2.5. The environment is fully reproducible via the provided conda environment and setup instructions in our repository.

4. Results

Our model achieves a validation MSE of 0.255 and EPE of 0.793m on the held-out shard after 25 epochs. The predicted flow vectors (Figure 1) align with the dominant ego-motion direction, consistent with the global flow prediction architecture. The loss curves (Figure 2) show rapid convergence within 10 epochs, followed by a plateau attributable to the small dataset size (822 samples).

5. Discussion

One limitation we identify is that the global flow head predicts one vector for all points, unable to capture spatially varying flow. The single-flight dataset limits diversity. These factors together explain the plateau observed after epoch 10.

Future works include: (1) Add PTv3 decoder for per-point flow prediction to capture spatially varying scene dynamics. (2) Combine real-world multi-flight and simulated drone data to reduce sensor noise. (3) Expand to a full workshop paper with baseline comparisons against FlowNet3D [3] and PointFlowNet [1].

6. Conclusion

We presented PTv3-Drone-Flow, adapting PointWorld’s PTv3-based world model from robot arm manipulation to drone ego-motion scene flow. The progression PointNet $\rightarrow$ PTv3 $\rightarrow$ PointWorld represents a coherent evolution — static per-point features, to local spatial attention, to temporal scene dynamics.

Our pipeline achieves val EPE of 0.793m on 822 real ROS2 samples, validating that PTv3-based world model can learn meaningful drone scene dynamics even at small scale.

References

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Author Contributions

Ray Chen — contributed to related work synthesis (PointWorld paper), designed & implemented the full experiment pipeline (ROS2 bag conversion, PTv3 training, visualization); conducted cloud GPU training; framed the final manuscript in conference format.

Jiajin Li — contributed to related work synthesis (PointNet paper) and paper writing; assisted with result interpretation.

Qingyong Meng — contributed to related work synthesis (PointLLM paper) and report writing; assisted with experimental analysis.

* All authors reviewed and approved final manuscript.